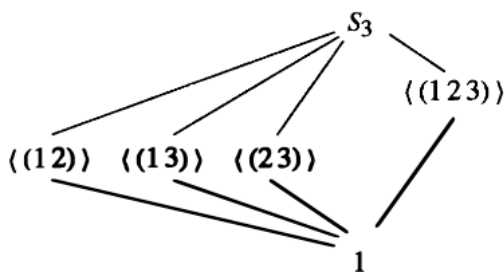


Section 2.5 - MY SOLUTIONS FROM GROUP HW

Exercise (2.5.6c). Use the given lattices to help find the centralizer of every element in S_3

FIGURE 1. Lattice Diagram for S_3 

Solution: Clearly $C_{S_3}(1) = S_3$. Seeing how the centralizer of an element $x \in S_3$ is a subgroup of S_3 containing x , by the way the lattice diagram is structured, $C_{S_3}(\langle x \rangle)$ must be equal to either $\langle x \rangle$ or S_3 .

Consider (12) and note that $(13)(12)(13) = (23)$, therefore $(13) \notin C_{S_3}(\langle (12) \rangle)$. Thus $C_{S_3}(\langle (12) \rangle) = \langle (12) \rangle$.

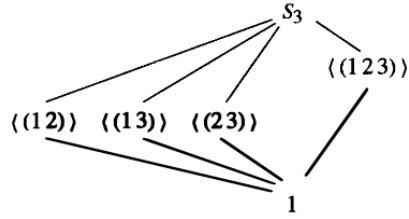
Consider (13) and note that $(12)(13)(12) = (23)$, therefore $(12) \notin C_{S_3}(\langle (13) \rangle)$. Thus $C_{S_3}(\langle (13) \rangle) = \langle (13) \rangle$.

Consider (23) and note that $(13)(23)(13) = (12)$, therefore $(13) \notin C_{S_3}(\langle (23) \rangle)$. Thus $C_{S_3}(\langle (23) \rangle) = \langle (23) \rangle$.

Consider (132) and note that $(13)(132)(13) = (123)$, therefore $(13) \notin C_{S_3}(\langle (132) \rangle)$. Since $(132) = (123)^2$, $C_{S_3}(\langle (132) \rangle) = C_{S_3}(\langle (123) \rangle) = \langle (123) \rangle$. $\square$

Exercise (2.5.8).

- (1) Find the normalizer of each subgroup in S_3 .

FIGURE 2. Lattice Diagram for S_3 

Solution: Clearly $N_{S_3}\langle 1 \rangle = S_3$ and $N_{S_3}\langle S_3 \rangle = S_3$. Similarly to the previous problem, since the normalizer of subgroup contains the subgroup, and because of the structure of the lattice diagram for any subgroup G , either $N_{S_3}(G) = G$ or $N_{S_3}(G) = S_3$.

Consider $\langle 12 \rangle$ and note that $(13)(12)(13) = (23)$. Since $(23) \notin \langle \langle(12)\rangle \rangle$ we know that $(13) \notin N_{S_3}(\langle \langle(12)\rangle \rangle)$ and therefore $N_{S_3}(\langle \langle(12)\rangle \rangle) = \langle \langle(12)\rangle \rangle$.

Consider $\langle 13 \rangle$ and note that $(12)(13)(12) = (23)$. Since $(23) \notin \langle \langle(13)\rangle \rangle$ we know that $(12) \notin N_{S_3}(\langle \langle(13)\rangle \rangle)$ and therefore $N_{S_3}(\langle \langle(13)\rangle \rangle) = \langle \langle(13)\rangle \rangle$.

Consider $\langle 23 \rangle$ and note that $(13)(23)(13) = (12)$. Since $(12) \notin \langle \langle(23)\rangle \rangle$ we know that $(13) \notin N_{S_3}(\langle \langle(23)\rangle \rangle)$ and therefore $N_{S_3}(\langle \langle(23)\rangle \rangle) = \langle \langle(23)\rangle \rangle$.

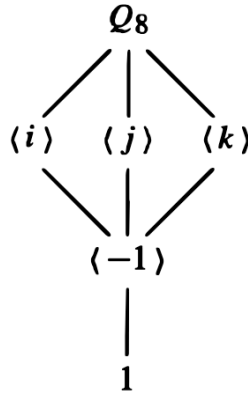
Finally consider $\langle \langle(123)\rangle \rangle = \{1, (123), (132)\}$, and note that

$$(12)(123)(12) = (132),$$

$$(12)(132)(12) = (123).$$

Therefore, $(12) \in N_{S_3}(\langle \langle(123)\rangle \rangle)$, and since $(12) \notin \langle \langle(123)\rangle \rangle$ we know that $N_{S_3}(\langle \langle(123)\rangle \rangle) = S_3$. $\square$

(2) Find the normalizer of each subgroup in Q_8

FIGURE 3. Lattice Diagram for Q_8 

Solution: Clearly $N_{Q_8}(\langle 1 \rangle) = Q_8$ and $N_{Q_8}(\langle Q_8 \rangle) = Q_8$. Let $a \in Q_8$ and note that, $a(-1)a^{-1} = a(-1)(-1)a = a^2 = -1 \in \langle -1 \rangle$. Thus $N_{Q_8}(\langle -1 \rangle) = Q_8$. Since the normalizer of subgroup contains the subgroup, and because of the structure of the lattice diagram for subgroups $G = \{\langle i \rangle, \langle j \rangle, \langle k \rangle\}$, either $N_{Q_8}(G) = G$ or $N_{Q_8}(G) = Q_8$.

Consider $\langle i \rangle$ and note that $ji(-j) = (-k)(-j) = -i \in \langle i \rangle$ and $j - i(-j) = (j)(k) = i \in \langle i \rangle$. Therefore $j \in N_{Q_8}(\langle i \rangle)$, thus $N_{Q_8}(\langle i \rangle) = Q_8$.

Consider $\langle j \rangle$ and note that $ij(-i) = (k)(-i) = -j \in \langle j \rangle$ and $i - j(-i) = (i)(-k) = j \in \langle j \rangle$. Therefore $i \in N_{Q_8}(\langle j \rangle)$, thus $N_{Q_8}(\langle j \rangle) = Q_8$.

Consider $\langle k \rangle$ and note that $ik(-i) = (-j)(-i) = -k \in \langle k \rangle$ and $i - k(-i) = (i)(j) = k \in \langle k \rangle$. Therefore $j \in N_{Q_8}(\langle k \rangle)$, thus $N_{Q_8}(\langle k \rangle) = Q_8$. $\square$

Section 3.1

Exercise (3.1.1). Let $\varphi : G \rightarrow H$ be a homomorphism and let E be a subgroup of H . Prove that $\varphi^{-1}(E) \leq G$. If $E \triangleleft H$ prove that $\varphi^{-1}(E) \triangleleft G$. Deduce that $\ker \varphi \triangleleft G$.

Proof: Suppose $\varphi : G \rightarrow H$ is a homomorphism and consider E , a subgroup of H . By properties of homomorphisms $1 \in \varphi^{-1}(E)$. Now consider $x, y \in \varphi^{-1}(E)$. By definition there exists some $a, b \in E$ such that $\varphi(x) = a$, and $\varphi(y) = b$. Consider xy and note that since φ is a homomorphism, $\varphi(xy) = \varphi(x)\varphi(y) = ab$. Since E is a group $ab \in E$ and therefore by definition $xy \in \varphi^{-1}(E)$. By properties of homomorphisms we know that $\varphi(x^{-1}) = \varphi(x)^{-1} = a^{-1}$, since $a^{-1} \in E$ we know that $x^{-1} \in \varphi^{-1}(E)$. Thus $\varphi^{-1}(E) \leq G$.

Suppose $E \triangleleft H$. By definition $heh^{-1} \in E$, for all $h \in H$ and $e \in E$. Let $g \in G$ and $a \in \varphi^{-1}(E)$. Note $\varphi(gag^{-1}) = \varphi(g)\varphi(a)\varphi(g)^{-1} = heh^{-1}$. Since $E \triangleleft H$, $heh^{-1} \in E$ and therefore $\varphi^{-1}(E) \triangleleft G$.

Let E be the trivial subgroup which is also normal in H . It follows that $\varphi^{-1}(E) = \ker \varphi$, and by the previous result $\ker \varphi \triangleleft G$. $\square$

Exercise (3.1.3). Let A be an abelian group and let B be a subgroup of A . Prove that A/B is abelian. Give an example of a non-abelian group G containing a proper normal subgroup N such that G/N is abelian.

Proof: Suppose A be an abelian group and let B be a subgroup of A . Consider $x, y \in A$, and note that $xB, yB \in A/B$. Since B is a subgroup of A we know that $xB yB = (xy)B$, and since A is abelian,

$$xB yB = (xy)B = (yx)B = yB xB.$$

Therefore A/B is abelian. $\square$

Maybe for an example consider Q_8 with normal subgroup $\langle -1 \rangle$.

Exercise (3.1.4). Prove that in the quotient group G/N , $(gN)^\alpha = g^\alpha N$ for all $\alpha \in \mathbb{Z}$.

Proof: Suppose N is a subgroup of G and consider the quotient group G/N . By Proposition 5 we know that, $g_1N \cdot g_2N = (g_1g_2)N$ is a well defined operation. What follows is that $(gN)^{-1} = g^{-1}N$. Therefore for $\alpha \in \mathbb{Z}$,

$$(gN)^\alpha = g_1N \cdot g_2N \cdot \dots \cdot g_\alpha N = (g_1g_2\dots g_\alpha)N = g^\alpha N.$$

□

Exercise (3.1.5). Use the preceding exercise to prove that the order of the element gN in G/N is n , where n is the smallest positive integer such that $g^n \in N$.

Proof: Suppose $gN \in G/N$, and let n be the smallest positive integer such that $g^n \in N$. By the previous problem we know that $(gN)^n = g^nN = N$ since $g^n \in N$. Thus $|gN| \leq n$. Suppose that $(gN)^i = N$, by definition it follows that $(gN)^i = g^iN = N$, therefore $g^i \in N$. Note that n is the smallest such integer such that and therefore $|gN| \geq n$. Thus $|gN| = n$. □

Exercise (3.1.7). Define $\pi : \mathbb{R}^2 \rightarrow \mathbb{R}$ by $\pi((x, y)) = x + y$. Prove that π is a surjective homomorphism and describe the kernel and fibers of π geometrically

Proof: Suppose $\pi : \mathbb{R}^2 \rightarrow \mathbb{R}$ by $\pi((x, y)) = x + y$. Let $(x, y), (a, b) \in \mathbb{R}^2$ and consider $\pi((x, y) + (a, b)) = \pi((a + x, b + y)) = a + x + b + y = a + b + x + y = \pi((a, b))\pi((x, y))$. Thus π is a homomorphism. Let $x \in \mathbb{R}$, consider $(x, 0) \in \mathbb{R}^2$ and note $\pi((x, 0)) = x + 0 = x$ and thus π is a surjection.

Now consider the kernel of π , and note that $\ker(\pi) = \{(x, -x) \in \mathbb{R}^2 | x \in \mathbb{R}\}$ since $\pi(x, -x) = x - x = 0$ which is the identity in $\mathbb{R}$. Geometrically this is the line $y = -x$. Recall that the fibers of π for a given element $a \in \mathbb{R}$ is the set of all $\phi^{-1}(a) = \{(x, y) \in \mathbb{R}^2 : x + y = a\}$. Thus fibers of π are of the form $y = a - x$, and geometrically represent lines parallel to the kernel. □

Exercise (3.1.22). a. Prove that if H and K are normal subgroups of group G then their intersection $H \cap K$ is also a normal subgroup of G .

Proof: Suppose H and K are normal subgroups of group G . Suppose $a, b \in H \cap K$ clearly $ab^{-1} \in H \cap K$ since H and K are subgroups and both contain a, b . Hence $H \cap K$ is a subgroup. Let $g \in G$ and $a \in H \cap K$ clearly $gag^{-1} \in H$ since H is normal in G , and $gag^{-1} \in K$ since K is normal in G . Thus $gag^{-1} \in H \cap K$ and therefore $H \cap K$ is a normal subgroup of G . $\square$

- b. Prove that the intersection of an arbitrary nonempty collection of normal subgroups of a group is a normal subgroup.

Proof: Suppose $\mathcal{H}$ is a family of nonempty normal subgroups in a group G . Consider the following $a \in \cap \mathcal{H}$. Clearly for $g \in G$, $gag^{-1} \in H_i$ for all $H_i \in \mathcal{H}$ since $\mathcal{H}$ is a family of normal subgroups. Thus by definition it follows that, $gag^{-1} \in \cap \mathcal{H}$. $\square$

Exercise (3.1.24). Prove that if N is a normal subgroup of G and H is any subgroup of G then $N \cap H$ is a normal subgroup of H .

Proof: Suppose N is a normal subgroup of G and H is any subgroup of G . Let $a, b \in N \cap H$, clearly $ab^{-1} \in N \cap H$, hence $N \cap H \leq H$. Note that since N is normal in G and H is a subgroup for all $h \in H$ $hah^{-1} \in N$. Since $a \in H$ and H is a subgroup we know that $hah^{-1} \in H$. Thus $hah^{-1} \in N \cap H$ and therefore $N \cap H$ is normal in H . $\square$

Exercise (3.1.28). Let N be a finite subgroup of a group G and assume $N = \langle S \rangle$ for some subset S of G . Prove that an element $g \in G$ normalizes N if and only if $gSg^{-1} \subseteq N$.

Proof: Let N be a finite subgroup of a group G and $N = \langle S \rangle$ for some subset S of G . Suppose $g \in G$ normalizes N . By definition it follows that for $g \in G$, $gng^{-1} \in N$. By definition N is also the intersection of all subgroups containing S , hence $S \subseteq N$ and since S is a subset of G for all $s \in S$, $gsg^{-1} \in N$. Thus $gSg^{-1} \subseteq N$.

Now suppose $gSg^{-1} \subseteq N$. Since N is the group generated by S by definition every n is equal to some unique, finite product of the elements of S , hence $n = s_1^{\alpha_1} s_2^{\alpha_2} \dots s_k^{\alpha_k}$.

Consider gng^{-1} and note that by substitution,

$$\begin{aligned}
 gng^{-1} &= gs_1^{\alpha_1} s_2^{\alpha_2} \dots s_k^{\alpha_k} g \\
 &= gs_1^{\alpha_1} (g^{-1}g) s_2^{\alpha_2} \dots (g^{-1}g) s_k^{\alpha_k} g^{-1} \\
 &= (gs_1^{\alpha_1} g^{-1})(gs_2^{\alpha_2} g^{-1}) \dots (gs_k^{\alpha_k} g^{-1}) \\
 &= (gs_1 g^{-1})^{\alpha_1} (gs_2 g^{-1})^{\alpha_2} \dots (gs_k g^{-1})^{\alpha_k}.
 \end{aligned}$$

Since $gSg^{-1} \subseteq N$ and N is a group, $gng^{-1} = (gs_1 g^{-1})^{\alpha_1} (gs_2 g^{-1})^{\alpha_2} \dots (gs_k g^{-1})^{\alpha_k} \in N$.

Thus $g \in G$ normalizes N . □

Exercise (3.1.31). Prove that if $H \leq G$ and N is a normal subgroup of H then $H \leq N_G(N)$. Deduce that $N_G(N)$ is the largest subgroup of G in which N is normal (i.e., is the join of all subgroups H for which $N \trianglelefteq H$)

Proof: Suppose $H \leq G$ and N is a normal subgroup of H . Consider $h \in H$, since N is a normal subgroup of H we know that $hNh^{-1} = N$ so clearly $h \in N_G(N)$. Therefore H is a subgroup contained in $N_G(N)$ and thus $H \leq N_G(N)$.

Consider the set B of all subgroups H of G such that $N \trianglelefteq H$. Note that by definition $N_G(N) \in B$. By the previous result every $H \in B$ has the property that $H \leq N_G(N)$ therefore $N_G(N)$ must be the largest subgroup of G normal to N . □

Exercise (3.1.35). Prove that $SL_n(F) \trianglelefteq GL_n(F)$ and describe the isomorphism type of the quotient group

Proof: Clearly the group $SL_n(F)$, the set of all $n \times n$ matrices with determinant 1 over the field F , it is contained within the set of all $n \times n$ matrices with nonzero determinant over the field F , hence $SL_n(F) \leq GL_n(F)$. Consider $X \in SL_n(F)$ and $Y \in GL_n(F)$ and note that by properties of determinant,

$$\det(YXY^{-1}) = \det(Y) \det(X) \det(Y^{-1}) = \det(Y)(1) \frac{1}{\det(Y)} = 1.$$

Therefore by definition $YXY^{-1} \in SL_n(F)$. Thus $SL_n(F) \trianglelefteq GL_n(F)$.

Consider the map $f : GL_n(F) \rightarrow (F^*)^x$ defined by $f(X) = \det(X)$. Let $A, B \in GL_n(F)$ and note that $f(AB) = \det(AB) = \det(A)\det(B) = f(A)f(B)$. By the first homomorphism theorem we know that $\ker(f) \trianglelefteq GL_n(F)$ and $G_n(F)/\ker(f) \cong f(GL_n(F))$. Note that since $A \in S_n(F)$ is defined as the set of matrices with $\det(A) = 1$. Thus $G/\ker(f) = G_n(F)/S_n(F)$. Also note that $f(GL_n(F)) = F^*$ by the definition of $GL_n(F)$, thus $G_n(F)/S_n(F) \cong F^*$. $\square$

Exercise (3.1.36 - Stefano Fochesatto). Prove that if $G/Z(G)$ is cyclic then G is abelian. [If $G/Z(G)$ is cyclic with generator $xZ(G)$, show that every element of G can be written in the form $x^a z$ for some integer $a \in \mathbb{Z}$ and some element $z \in Z(G)$.]

Proof: Suppose $G/Z(G) = \langle xZ(G) \rangle$ for some fixed $x \in G$. Since $G/Z(G)$ is cyclic there exists some $a \in \mathbb{Z}$ such that for all $g \in G$,

$$gZ(G) = (xZ(G))^a = x^a Z(G).$$

Now consider $a, b \in G$, and note since $Z(G)$ is a subgroup of G we know $G/Z(G)$ forms a partition of G and thus a and b belong to some left coset of $Z(G)$. Therefore by the previous result there exists some $\alpha, \beta \in \mathbb{Z}$ and $z_1, z_2 \in Z(G)$ such that $a = x^\alpha z_1$ and $b = x^\beta z_2$. Since z_1, z_2 are commute with every element in G ,

$$ab = x^\alpha (z_1 x^\beta z_2) = x^{\alpha+\beta} (z_1 z_2) = x^{\alpha+\beta} z_2 z_1 = x^\beta (x^\alpha z_2) z_1 = x^\beta z_2 x^\alpha z_1 = ba$$

$\square$

Exercise (3.1.37). Let A and B be groups. Show that $\{(a, 1) | a \in A\}$ is a normal subgroup of AxB and the quotient of AxB by this subgroup is isomorphic to B .

Proof: Suppose A and B are groups. Let $X = \{(a, 1) | a \in A\}$, and consider $(a, b) \in AxB$, and $(x, 1) \in X$. Note that $(a, b)(a, 1)(a, b)^{-1} = (a, b)(a, 1)(a^{-1}, b^{-1}) = (axa^{-1}, b1b^{-1}) = (axa^{-1}, 1)$. Since A is a subgroup and $x \in A$, $axa^{-1} \in A$ and therefore $(axa^{-1}, 1) \in X$.

Thus $X \trianglelefteq A$.

Consider AxB/X , and define the mapping $\varphi : AxB/X \rightarrow B$ by $\varphi((a, b)X) = b$.

Let $(a, b)X = (\alpha, \beta)X$. Therefore it follows that the product $(a, b)(\alpha, \beta)^{-1} = (a, b)(\alpha^{-1}, \beta^{-1}) = (a\alpha^{-1}, b\beta^{-1}) \in X$. Thus $b\beta^{-1} = 1$ and $b = \beta$ so φ is well defined.

Note that $\varphi((a, b)X(\alpha, \beta)X) = \varphi((a, b)(\alpha, \beta)X) = \varphi((a\alpha, b\beta)X) = b\beta = \varphi((a, b)X)\varphi((\alpha, \beta)X)$.

Thus φ is a homomorphism.

Let $\varphi((a, b)X) = \varphi((\alpha, b)X)$. Note that $(a, b)(\alpha b)^{-1} = (a\alpha, bb^{-1}) = (a\alpha, 1)$. By definition $(a\alpha, 1) \in X$ so $(a, b)X = (\alpha, b)X$. Thus φ is injective. Clearly φ is a surjection, let $b \in B$ since the domain is AxB/X we can always construct an $(a, b)X$ such that $\varphi((a, b)X) = b$. $\square$

Exercise (3.1.40). Let G be a group, let N be a normal subgroup of G and let $\overline{G} = G/N$. Prove that $\overline{x}$ and $\overline{y}$ commute in $\overline{G}$ if and only if $x^{-1}y^{-1}xy \in N$.

Proof: Let G be a group, $N \trianglelefteq G$, and $\overline{G} = G/N$. Suppose for $\overline{x}, \overline{y} \in \overline{G}$, $\overline{xy} = \overline{yx}$. It follows that $xyN = yxN$ and for some $n_1, n_2 \in N$ we know that,

$$xyn_1 = yxn_2,$$

$$(yx)^{-1}xyn_1 = n_2,$$

$$x^{-1}y^{-1}xy = n_2n_1^{-1}.$$

Since N is a subgroup we know that $x^{-1}y^{-1}xy \in N$.

Now suppose that $x^{-1}y^{-1}xy \in N$. Let $n \in N$ and consider the following,

$$x^{-1}y^{-1}xy = n$$

$$xy = yxn$$

Therefore $xy \in yxN$. Since N is normal the cosets of G/N partition G , and since $xy \in yxN$ it must be the case that $xyN = yxN$. $\square$

Exercise (3.1.41). Let G be a group. Prove that $N = \langle x^{-1}y^{-1}xy | x, y \in G \rangle$ is a normal subgroup of G . and G/N is abelian.

Proof: Let G be a group, and suppose that $N = \langle x^{-1}y^{-1}xy | x, y \in G \rangle$. Consider $g \in G$. Note that for the generator of N , $gx^{-1}y^{-1}xyg^{-1} = gx^{-1}(g^{-1}g)y^{-1}(g^{-1}g)x(g^{-1}g)yg^{-1} = (gx^{-1}g^{-1})(gy^{-1}g^{-1})(gxxg^{-1})(gyyg^{-1}) = (gxxg^{-1})^{-1}(gyyg^{-1})^{-1}(gxxg^{-1})(gyyg^{-1})$. Since G is a group $gxxg^{-1}, gyyg^{-1} \in G$ therefore by definition $gx^{-1}y^{-1}xyg^{-1} \in N$. Thus $N \trianglelefteq G$. Since $x^{-1}y^{-1}xy \in N$ for all $x, y \in G$, and $N \trianglelefteq G$ by the previous exercise G/N must be abelian. $\square$

Exercise (3.1.42). Assume both H and K are normal subgroups of G with $H \cap K = 1$. Prove that $xy = yx$.

Proof: Suppose both H and K are normal subgroups of G and $H \cap K = 1$. Let $x \in H$ and $y \in K$. Consider the element $x^{-1}y^{-1}xy$. Note that since $x \in H$, $y \in G$, and $H \trianglelefteq G$ we know that there exists some $x_1 \in H$ such that $x = yx_1y^{-1}$. By substitution we get, $x^{-1}y^{-1}(yx_1y^{-1})y = x^{-1}x_1 \in H$. Now note that since $y^{-1} \in K$, $x \in G$, and $K \trianglelefteq G$ there exists some $y_1 \in K$ such that $y^{-1} = xy_1x^{-1}$. By substitution we get that $x^{-1}y^{-1}xy = x^{-1}(xy_1x^{-1})xy = y_1y \in K$. Therefore it follows that $x^{-1}y^{-1}xy \in H \cap K$. Thus $x^{-1}y^{-1}xy = 1$ and clearly $xy = yx$. $\square$

Section 3.2

Exercise (3.2.4). Show that if $|G| = pq$ for some primes p and q then either G is abelian or $Z(G) = 1$

Proof: Consider G such that $|G| = pq$ for some primes p and q . Since $|G| = pq$ By Lagrange's Theorem it follows that $|Z(G)| = 1, p, q, pq$. Clearly if $|Z(G)| = 1$ then $Z(G) = 1$ and we are done, similarly if $|Z(G)| = pq$ then $Z(G) = G$ and therefore G is abelian.

Without loss of generality let $|Z(G)| = p$. By Lagrange's Theorem $|G/Z(G)| = |G|/|Z(G)| = pq/p = q$. Since $|G/Z(G)|$ is of prime order then $G/Z(G)$ must be cyclic. By 3.1.36 if $G/Z(G)$ is cyclic then G is abelian. $\square$

Exercise (3.2.5). Let H be a subgroup of G and fix some element $g \in G$.

a Prove that gHg^{-1} is a subgroup of G of the same order as H .

Proof: Suppose H is a subgroup of G and fix some element $g \in G$. Consider the group gHg^{-1} . Clearly this group is non empty, as it contains the identity. Let $a, b \in gHg^{-1}$ for some $h_a, h_b \in H$ such that $a = gh_ag^{-1}$ and $b = gh_bg^{-1}$. Note that $ab^{-1} = gh_ag^{-1}gh_b^{-1}g^{-1} = gh_a h_b^{-1}g^{-1}$. Since $H \leq G$ we know that $h_a h_b^{-1} \in H$ thus $ab^{-1} \in gHg^{-1}$ and $gHg^{-1} \leq G$.

To show that $|gHg^{-1}| = |H|$ we will prove they are isomorphic. Consider the map $\varphi : H \rightarrow gHg^{-1}$ defined by $\varphi(h) = ghg^{-1}$. Note that this map is a homomorphism, for all $a, b \in H$ we know that $\varphi(ab) = gabg^{-1} = gag^{-1}gbg^{-1} = \varphi(a)\varphi(b)$. This map is also an injection, if $\varphi(a) = \varphi(b)$ then we know that $gag^{-1} = gbg^{-1}$, and thus $a = b$. The map is a surjection by definition. Thus $gHg^{-1} \cong H$ and therefore $|gHg^{-1}| = |H|$. $\square$

b Deduce that if $n \in \mathbb{Z}^+$ and H is the unique subgroup of G of order n then $H \trianglelefteq G$.

Proof: Suppose that H is the unique subgroup of G of order $n \in \mathbb{Z}^+$. Note that by the previous exercise the subgroup gHg^{-1} must have the same order of H , since H has unique order we know $gHg^{-1} = H$. Thus $H \trianglelefteq G$. $\square$

Exercise (3.2.8). Prove that if H and K are finite subgroups of G whose orders are relatively prime then $H \cap K = 1$

Proof: Suppose H and K are finite subgroups of G such that $|H|$ and $|K|$ are relatively prime. Consider the group $H \cap K$. Clearly since H and K are groups $H \cap K \leq H, K$. By Lagrange's Theorem $|H \cap K| \mid |H|$ and $|H \cap K| \mid |K|$. Since $|H|$ and $|K|$ are

relatively prime the only common divisor is 1, therefore it must be the case that $|H \cap K| = 1$. Thus $H \cap K = 1$. $\square$

Exercise (3.2.10). Suppose H and K are subgroups of finite index in the group G with $|G : H| = m$ and $|G : K| = n$. Prove that $\text{lcm}(m, n) \leq |G : H \cap K| \leq mn$. Deduce that if m and n are relatively prime then $|G : H \cap K| = |G : H||G : K|$.

Proof: Let H and K are subgroups of finite index in the group G . Suppose that $|G : H| = m$ and $|G : K| = n$. Consider the set of left cosets of $H \cap K$ in G , denoted by $|G : H \cap K|$. Let $a \in g(H \cap K)$ and note that by definition $a \in gH$ and $a \in gK$. Now consider the left cosets of H and K in G . Clearly if $a \in gH$ and $a \in gK$ by left cancellation $g^{-1}a \in H \cap K$, and thus $a \in g(H \cap K)$. Therefore $gH \cap gK = g(H \cap K)$. Thus every left coset of $(H \cap K)$ is defined by the intersection of some left coset of H and some left coset of K . Seeing as there are m left cosets of H and n left cosets of K there are at most mn left cosets of $(H \cap K)$.

By Lagrange's Theorem it follows that $|G : H \cap K| = |G : H||H : H \cap K|$. Thus it follows that $m ||G : H \cap K|$. By the same argument we get that $n ||G : H \cap K|$. Therefore the minimum $|G : H \cap K|$ can be is the $\text{lcm}(m, n)$. Thus $\text{lcm}(m, n) \leq |G : H \cap K| \leq mn$.

When m and n are relatively prime we know that $\text{lcm}(m, n) = mn$ and therefore very clearly $|G : H \cap K| = mn = |G : H||G : K|$. $\square$

Exercise (3.2.16). Use Lagrange's Theorem, in the multiplicative group $(\mathbb{Z}/p\mathbb{Z})^x$ to prove Fermat's Little Theorem: if p is prime then $a^p \cong a \pmod{p}$ for all $a \in \mathbb{Z}$.

Proof: Suppose p is prime and let $a \in \mathbb{Z}$. If a is a multiple of p then clearly $a^p \cong 0 \cong a \pmod{p}$. If a is not a multiple of p then its residual α belongs to some residual class in $(\mathbb{Z}/p\mathbb{Z})^x$. Recall that since p is prime $|(\mathbb{Z}/p\mathbb{Z})^x| = \varphi(p) = p - 1$ where $\varphi(p)$

denotes Euler's φ -function By Lagrange's Theorem (Corollary 9) we know that for $\alpha \in (\mathbb{Z}/p\mathbb{Z})^x$, $\alpha^{|\mathbb{Z}/p\mathbb{Z}|} = 1$. Hence $a^p = a^{p-1}a \cong \alpha^{p-1}a \pmod{p} \cong a \pmod{p}$. $\square$

Exercise (3.2.22). Use Lagrange's Theorem, in the multiplicative group $(\mathbb{Z}/n\mathbb{Z})^x$ to prove Euler's Theorem: $a^{\varphi(n)} \cong 1 \pmod{n}$ for every integer a relatively prime to n , where φ denotes Euler's φ -function.

Proof: Consider the multiplicative group $(\mathbb{Z}/n\mathbb{Z})^x$ and suppose $a \in \mathbb{Z}$ such that a and n are relatively prime. Recall that for a multiplicative group modulo n , $|(\mathbb{Z}/n\mathbb{Z})^x| = \varphi(n)$. Since a is relatively prime to n we know that $a \in (\mathbb{Z}/n\mathbb{Z})^x$. Thus by Lagrange's Theorem (Corollary 9) we know that $a^{\varphi(n)} \cong 1 \pmod{n}$. $\square$

Exercise (3.2.23). Determine the last two digits of $3^{3^{100}}$. [Determine $3^{100} \pmod{\varphi(100)}$ and use the previous exercise.]

Solution: Computing $\varphi(100)$, since $100 = 2^2 5^2$ we get ,

$$\varphi(100) = \varphi(2^2)\varphi(5^2) = (2^2 - 2)(5^2 - 5) = 40.$$

By the previous result we know that $3^{\varphi(40)} \cong 1 \pmod{40}$. Since $40 = 2^3 5$ we know that $\varphi(16) = (2^3 - 2^2)(5 - 1) = 16$. Therefore $3^{16} \cong 1 \pmod{40}$. It follows that $3^{100} = 3^{6(16)+4} \cong 3^4 \pmod{40} \cong 81 \pmod{40} \cong 1 \pmod{40}$. Thus $3^{100} = 1 + 40i$ for some $i \in \mathbb{Z}$. By substitution we get the following,

$$3^{3^{100}} = 3^{1+40i} = 3^1 3^{40i} = 3 * 3^{\varphi(100)i}.$$

Applying the previous result again we get that,

$$3^{3^{100}} = 3 * 3^{\varphi(100)i} \cong 3 \pmod{100}.$$

Thus the last two digits of $3^{3^{100}}$ are 03. $\square$